Muhammad Khaquan

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Education

University of Michigan, Ann Arbor Sep 2026 - May 2028 (expected)
M.S. in Computer Science and Engineering | **Coursework:** ML for Engineering

Lahore University of Management Sciences (LUMS) Sep 2021 - Jun 2025
B.S. in Computer Science; **GPA: 3.72/4.00**; Distinction; Dean's Honour List
Relevant coursework: Advanced Topics in ML, Generative AI, Machine Learning, Computer Vision Fundamentals, Convex Optimization, Applied Probability

Technical Skills

Languages / Tools: Python, C++, C, Git, OpenAI API, Codex (AI-assisted development)
Libraries: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, NumPy
Applied AI: Large Language Models (LLMs), Vision-Language Models, AI Agents, Deep Learning
Engineering: Data Structures, Algorithms, Object-Oriented Programming, Debugging, Regression Testing

Research Experience

GameScientist | **Research Assistant**, Touchstone AI Lab Aug 2026 - Present
- Developed **Balatro integration and native macOS interaction** for a vision-language model benchmark using Python, PyObjC, ScreenCaptureKit, and Quartz; added 2 tasks and validated 1280x720 capture and mouse/keyboard actions.
- **Defined requirements and guided technical decisions**; directed **Codex-assisted implementation and debugging**, evaluating proposed changes through observed behavior, corrective feedback, and regression testing.
- Built screenshot-gated evaluation for **3 ante checkpoints** with **21 checkpoint/provider regression tests**; distinguished live operator checks, offline tests, and model-driven gameplay when assessing results.

ReasonDistill | **Research Lead**, CSALT + City@LUMS Oct 2025 - Present
- Improved **LLM reasoning** over baseline knowledge distillation on ARC-C, GSM8K, MMLU, MATH-500, and HellaSwag by transferring capabilities from Qwen-3.5-4B to Qwen-3.5-2B.
- Lead four researchers in mechanistic distillation; trained a **sparse autoencoder** to identify reasoning-relevant structures and guide transfer to smaller language models.

AENet | **Research Lead**, City@LUMS Jul 2025 - Mar 2026
- Improved accuracy by **10-20 percentage points** over baselines on selected VLCS/OfficeHome domains unseen by the student using autoencoder-compressed teacher features; the teacher saw all domains.
- Directed **experimental design and evaluation** against logits-based distillation and FitNets variants; investigated transformation invariance, teacher-size effects, and latent sparsity.

Dynamic Grouped Query Attention | **Research Intern**, CSALT Aug 2024 - Dec 2024
- Increased ViT-L accuracy from **67.73% to 75.83%** versus grouped-query attention on Tiny ImageNet by adapting query groups to key-head norms; evaluated three model sizes across four datasets.
- **Co-first authored** *Beyond Uniform Query Distribution: Key-Driven Grouped Query Attention* (arXiv preprint).

Professional Experience

WhichDraft | **Machine Learning Engineer (Contract)** Jun 2023 - Jan 2024
- Built and shipped an **AI agent** for contract creation and editing using the **OpenAI API**, automating 90%+ of contract-taxonomy generation; tested correctness across 12 legal-document classes.
- Reduced formatting errors by over **60%** on an isolated suite of 100 challenging outputs through **prompt engineering**, debugging agent responses, and storing accepted formats in agent memory.

Environ AI | **Machine Learning Engineer (Contract)** Jan 2023 - Feb 2023
- Optimized **semantic search over 1M+ documents** using semantic hashing, increasing retrieval throughput by **42%**; benchmarked mean latency across 200+ synthetic queries using Python's time module.

LUMS Computer Science Department | **Teaching Fellow** Aug 2025 - Jul 2026
- Taught **Data Structures and Object-Oriented Programming** to 300-400 students per class; designed 14 **C/C++** labs and assessments, and coordinated teams of up to 30 teaching assistants.